Assignment-3.3

TASK-1

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

1. CMOS Battery Low / CMOS Checksum Bad

Issue: CMOS battery may be weak, or BIOS settings may be corrupted/reset.

2. Keyboard Error / No Keyboard Present

Issue: Keyboard may be disconnected, faulty, or the USB/PS/2 port may have a problem.

3. Disk Boot Failure / Insert System Disk and Press Enter

Issue: HDD/SSD may not be detected, disconnected, or the boot drive/order may be incorrectly configured.

4. Memory Test Fail / Decreasing Memory Size

Issue: RAM may be loose, faulty, or incompatible with the motherboard.

5. CPU Fan Error / Fan Failed

Issue: CPU cooling fan may be disconnected, not spinning, or incorrectly connected to the CPU_FAN header.

TASK-2

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

1. Reseat the RAM Modules

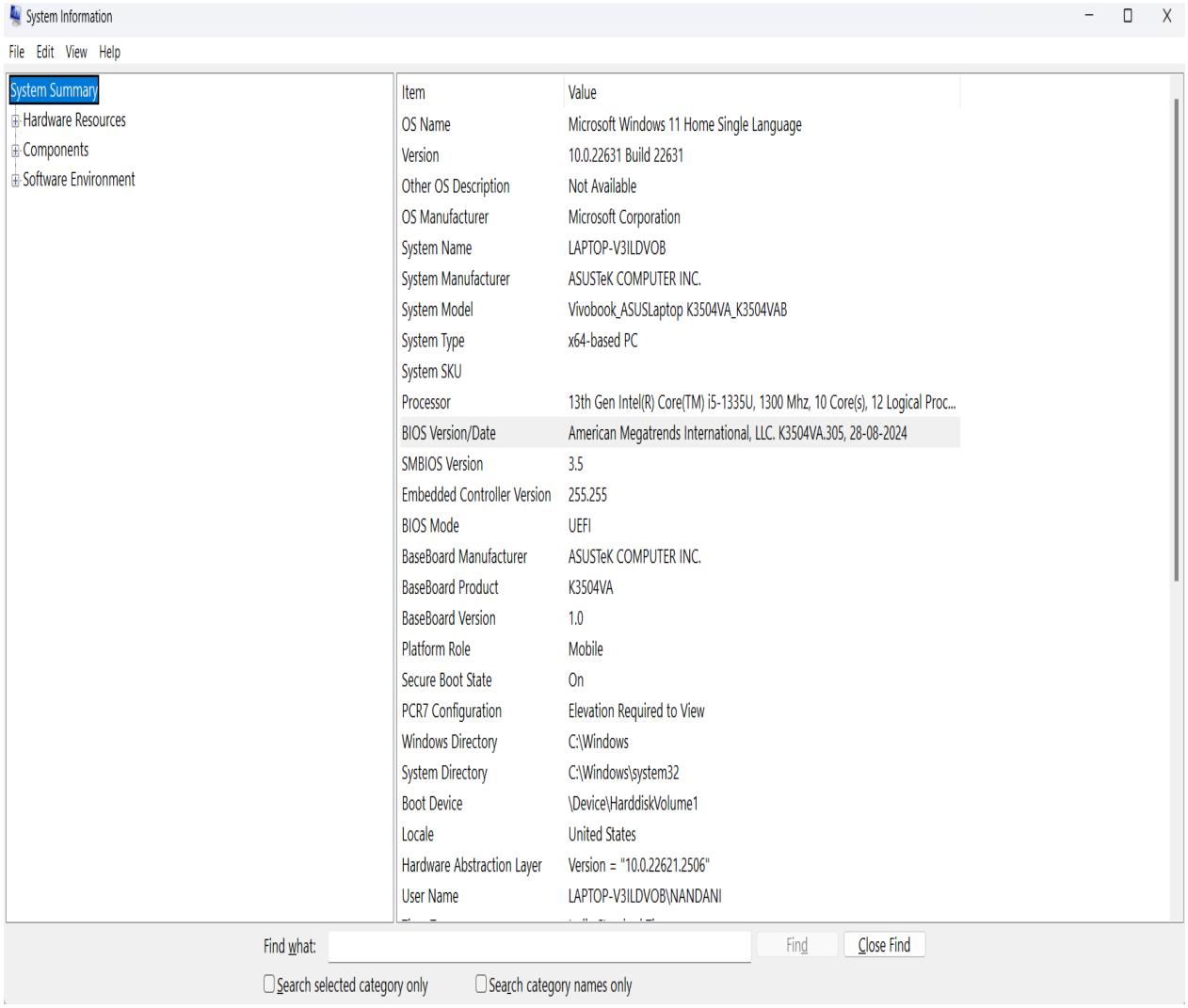
- Shut down the computer completely, turn off the power supply, and unplug the power cable.
- Open the PC case, locate the RAM slots on the motherboard, and press the side tabs to release the RAM sticks.
- Clean any dust out of the slots. Firmly push the RAM sticks back into the slots until the side clips click shut to ensure a secure connection. Try booting again.

2. Isolate a Faulty Module or Slot

- If you have multiple RAM sticks, remove all of them except for one stick in the primary slot (usually slot 1/A1).
- Try booting the PC. If it boots successfully, the stick you removed is likely defective. If it still beeps, swap it with the other stick to test it individually.
- If both sticks give the same error in that specific slot, try testing a known working stick in a different slot to rule out a faulty motherboard channel.

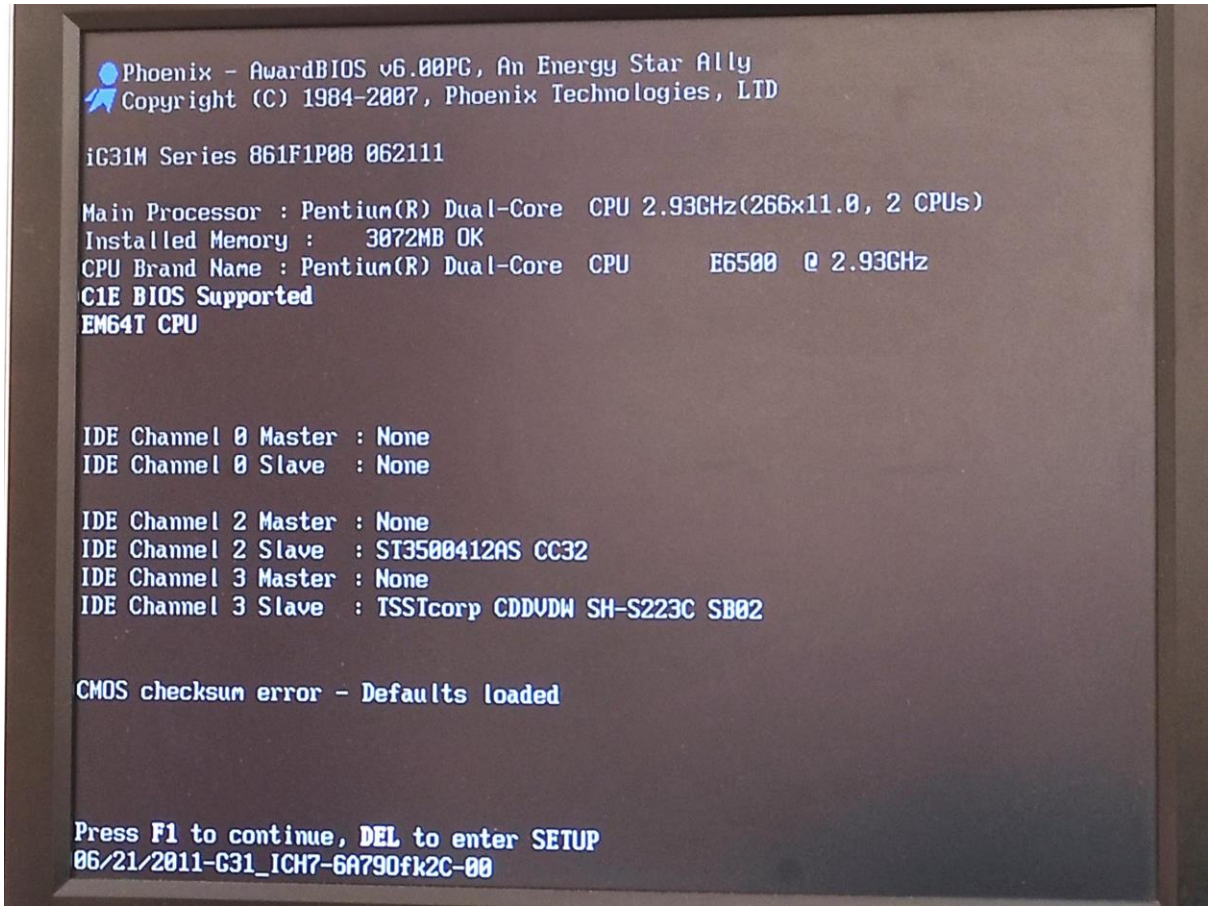
TASK-3

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



TASK-4

You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.



1. Enter the BIOS/UEFI setup and check the date, time, and other settings.

2. Choose Load Optimized/Default Settings, then save and restart.

3. If the error appears again, turn off and unplug the PC.

4. Replace the CMOS battery.

5. Start the PC, enter BIOS again, set the correct date/time and boot settings, and save.

6. If the error continues even after replacing the battery, check for BIOS corruption or a motherboard/CMOS circuit problem.